

SOLUTION

Introduction:->

A solution is a homogeneous mixture of two or more chemically non-reacting substance whose composition can be varied within certain limits.

Solution may also be called single phase which contains more than one components.

→ A binary solution is made up of two components. Similarly, the solution made of three and four components are called ternary and quaternary solutions respectively.

→ The component which is present in larger amount in solution is called solvent and the other which is present in lesser amount is called solute.

* Types Of solutions:->

1. Gaseous solution:-> when a gas is mixed with other gas, then gaseous solution is obtained.

All gaseous mixtures are homogeneous solutions.

2. Liquid solution:-> when gas, liquid or solid is dissolved in liquid, then liquid solution is obtained.

3. Solid solutions:-> when gas, liquid or solid is mixed with other solid irregularly, then a solid solution is obtained.

→ On the basis of physical state of solute & solvent, solution can be classified into 3 different types:

Type of Solution	Solvent	Solute	Example
1. Gaseous solution	Gas	Gas Liquid Solid	Mixture of 2 gases, air water vapours in air sublimation of a solid into gas e.g. Camphor, NH_4Cl
2. Liquid solution	Liquid	Gas Liquid Solid	Soda water, ammonical water Mixture of miscible liquids Salt in water, sugar in water
3. Solid solution	Solid	Gas Liquid Solid	Adsorption of gases over metals. e.g. Adsorption of H_2 over Pd Hg in Cu , Hg in Zn Alloy Cu-Au , Zn-Cu

Determination of concentration of the solution :-

The quantity of solute present in certain amount of solvent is called concentration of solution. The concentration of solution can be expressed in various common ways as discussed below:

1. Mass percentage (w/w) → The percentage by mass means the mass of the solute in grams present in 100g of the solution.

$$\begin{aligned}
 \text{Mass percentage} &= \frac{\text{Amount of solute (in g)}}{\text{Amount of solution (in g)}} \times 100 \\
 &= \frac{\text{Amount of solute (in g)}}{\text{Amount of solute (in g)} + \text{Amount of solvent (in g)}} \times 100
 \end{aligned}$$

→ Generally, solute is represented by B and solvent is represented by A - $\therefore \% \text{ w/w} = \frac{w_B}{w_B + w_A} \times 100$

Example → 500 ml solution is formed by 11 g oxalic acid. The density of this solution is 1.1 g mL^{-1} . Determine the mass percentage of oxalic acid in the solution.

Ans. Mass of oxalic acid = 11 g

$$\begin{aligned}\text{mass of solution} &= 500 \text{ mL} \times 1.1 \text{ g mL}^{-1} \\ &= 550 \text{ g}\end{aligned}$$

$$\begin{aligned}\text{mass percentage of oxalic acid} &= \frac{11}{550} \times 100 \\ &= 2\% \text{ w/w}\end{aligned}$$

2. Volume percentage:-(V/V)

This method is used in the determination of conc. of solution having solute & solvent in liquid state.

The percentage by volume means the mass of the solute dissolved in 100 cm^3 or 100 mL of the solution.

$$\text{Volume percentage} = \frac{\text{Volume of solute (in mL)}}{\text{Volume of solution (in mL)}} \times 100$$

$$= \frac{\text{Volume of solute (in mL)}}{\text{Volume of solute (mL)} + \text{Volume of solvent (mL)}} \times 100$$

$$\left[\% \text{ V/V} = \frac{V_B}{V_A + V_B} \times 100 \right]$$

where-

V_B = Volume of solute & V_A = Volume of solvent (mL)

Example → The aqueous solution of 25% (V/V) of ethanol means that 25 ml ethanol is present in 75 ml water and the total volume of the solution is 100 ml.

$$\left\{ \text{Mass-Volume percentage} = \frac{\text{Amount of solute (in g)}}{\text{Amount of solution (in mL)}} \times 100 \right\}$$

Example → How many grams of NaCl will be required to make 500ml aqueous solution of 2% w/v NaCl?

Ans -

$$\text{Mass-Volume \%} = \frac{\text{Amount of solute (in g)}}{\text{Volume of solution (in mL)}} \times 100$$

$$2 = \frac{W_B}{500} \times 100$$

$$W_B = 10 \text{ g (Mass of solute - NaCl)}$$

4. Parts per million (PPm):- When a solute is present in very minute amounts. Its conc. is expressed in PPm.

$$\left\{ \text{PPm} = \frac{\text{mass of solute}}{\text{mass of solution}} \times 10^6 \right\}$$

5. Normality (N) - It is defined as the number of g-equivalents of the solute present in one litre of the solution.

$$\left\{ \text{Normality (N)} = \frac{\text{g-equivalents of solute}}{\text{Volume of solution in Ltr}} \right\}$$

→ A solution having normality equal to unity is called a normal solution.

6. Molarity (M):- It is defined as the number of gram mole, of the solute present in 1 Lt. of the solution.

$$\left\{ \text{Molarity (M)} = \frac{\text{moles of solute}}{\text{Volume of solutions in Lt}} \right\}$$

→ A solution having molarity equal to 1 unity is called molar solution.

Both normality and molarity of a solution change with change in temperature.

7. Molality (m):- It is defined as the number of gram mole of the solute present in 1000gm (1kg) of the solvent.

$$\left\{ \text{Molality (m)} = \frac{\text{mole of solute}}{\text{mole of solvent (kg)}} \right\}$$

→ A solution containing one mole of solute per 1000gm. of solvent has molality equal to 1, and is called a molal solution.
The molality of a solution does not change with temperature.

∴ Relation B/w molarity and normality:-

→ (Molarity $\times$ molecular mass of solute = Normality $\times$ Equivalent mass of solute)

For acid: Normality = Molarity $\times$ Basicity

For Base: Normality = Molarity $\times$ Acidity.

→ Basicity is the number of H^+ ions furnished by each molecule of acid in solution.

→ Acidity is the number of OH^- ions furnished by each molecule of base in solution.

∴ Relation B/w molarity (M) & mass percentage:-

$$* \text{ Molarity} = \frac{\% \text{ mass} \times d \times 10}{\text{molar mass of solute}} \quad (d = \text{density})$$

$$* \text{ Normality} = \frac{\% \text{ mass} \times d \times 10}{\text{equivalent mass of solute}}$$

∴ Dilution formula → If the solution of same substance is diluted by adding solvent from Volume V_1 to Volume V_2 then initial molarity M_1 and final molarity M_2 are related as -
 $[M_1 \times V_1 = M_2 \times V_2]$

∴ Molarity of a mixture → If V_1 ml of a solution of molarity M_1 is mixed with V_2 ml of another solution of same

Substance with molarity M_2 then molarity of the resulting mixture solution can be obtained by -

$$\left[M = \frac{M_1 V_1 + M_2 V_2}{V_1 + V_2} \right]$$

∴ Mole fraction → It is defined as the ratio of number of moles of one component to the total number of moles of all the components present in the solution.

Let us suppose that a solution contains the component A and B and suppose that W_A of A and W_B of B are present in it.
then -

$$\text{moles of A } (n_A) = \frac{W_A}{\text{molar mass of A}}$$

$$\text{moles of B } (n_B) = \frac{W_B}{\text{molar mass of B}}$$

$$\text{Total no. of moles} = n_A + n_B$$

$$\text{mole fraction of A} = \frac{n_A}{n_A + n_B} = x_A$$

$$\text{mole fraction of B} = \frac{n_B}{n_A + n_B} = x_B$$

→ Now, the sum of the moles fraction of solute & solvent in binary solution is unity.

$$x_A + x_B = \frac{n_A}{n_A + n_B} + \frac{n_B}{n_A + n_B} = \frac{n_A + n_B}{n_A + n_B} = 1$$

→ mole fraction is independent of the temperature.

∴ Relation B/w molality (m) and mole fraction of solute -

$$x_B = \frac{m \times \text{molar mass of A}}{1000 + m \times \text{molar mass of A}} \quad (\because \text{Molar mass of A} = MM_A)$$

$$\left\{ m = \frac{100 \times x_B}{x_A \times (MM_A)} \right\}$$

* Relation B/w Molarity (M) and mole fraction →

$$x_B = \frac{M \times (MMA)}{(MMA - MM_B) + 1000d}$$

$$\left\{ M = \frac{1000d \times x_B}{x_A \times (MMA) + x_B \times (MM_B)} \right\}$$

→ where A is solvent and B is solute.

÷ Formality :- It is defined as the number of gram formal mass (moles) of the ionic solute present in one litre of the solution.

It is represent by 'F'.

$$\left\{ F = \frac{\text{number of moles of ionic solute}}{\text{Volume of solution in litre}} \right\}$$

→ Formality of a solution change with change in temperature.

Solubility → The maximum amount of solid (in g) that can be dissolve in 100gm. liquid solvent at certain temperature is called solubility of solid.

The solution is called saturated solution in this stage.

* Factors affecting solubility of solids →

1. Nature of solid solute and solvent → Ionic solids are soluble in polar solvent -

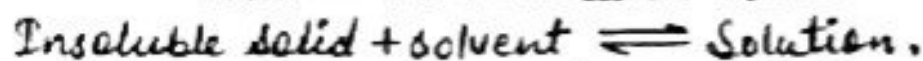
e.g. - NaCl, KCl, KNO₃ etc. These are solution in water. Its main reason is inter-ionic dipolar interactions.

The solubility of various ionic solids are different in water.

→ The ionic solids whose lattice energy is more than hydration energy, are insoluble in water while those solids whose lattice E is lesser than hydration energy are soluble in water.

Non-polar solids (e.g. I₂, S₂ etc.) are soluble in non-polar solvents like CCl₄, CS₂ etc.

2. Temperature:- There is an equilibrium b/w insoluble solid and solution in saturated solution:



$\Delta H = +x \text{ kcal}$. According to Le-Chatelier's principle, if $\Delta H > 0$, then heat is absorbed on dissolving solute at this condition, the solubility of solid solute will increase on increase in temperature.
e.g. NH_4Cl , KCl , AgNO_3 , NaNO_3 , KI etc.

If $\Delta H < 0$ then heat is released on dissolving solute in solvent. So, the solubility of solid solute will decrease with increase in temperature.

e.g. NaOH , Li_2SO_4 , $(\text{CH}_3\text{COO})_2\text{Ca}$ etc.

→ Some salts show abnormal behaviour. For example, the solubility of Glauber salt ($\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$) increases till 32.4°C , but the solubility decreases at higher temperature.

Its main reason is the separation of water of crystallization of this salt at 32.4°C and it becomes anhydrous salt.

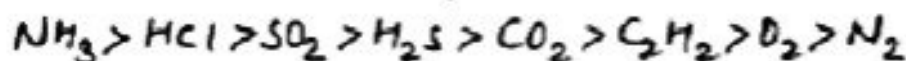
* Solubility of gases in liquids:

All gases are less or more soluble in water or soluble to a certain limit and form true solution.

The volume (in mL) of a gas which dissolved in 1 mL water is called absorption coefficient. The solubility of gases in liquids is affected by following factors:

1. Nature of gas: The gases which react with solvent or ionised in solution, are soluble in water, e.g. NH_3 , HCl(g) , SO_2 gases are more soluble in water and form NH_4OH , HCl(l) and H_2SO_4 compounds respectively.

The decreasing order of absorption coefficient of various gases in 1 mL water is given below:



→ N_2 , O_2 and CO_2 are more soluble in ethyl alcohol than water.

→ O_2 is readily soluble in blood.

2. Nature of solvent → The polar gases, are soluble in polar solvents while non-polar gases are soluble in non-polar solvents.

It is like dissolves like rule e.g. HCl gas is more soluble in water than liquid benzene.

3. Effect of temperature → According to Le-Chatelier's principle, at constant pressure, the solubility of gases decreases with increase in temperature.

Since the transformation kinetic energy of gaseous molecules in liquid increases on increasing temperature and hence on heating escaping tendency of gaseous molecules increases.

Exception → Heat of absorbed on dissolving H_2 and He in liquid. Hence the solubility of these gases increase with increasing temperature.

4. Effect of pressure → The solubility of gases is more affected by change in pressure.

Henry proposed a relationship between pressure (P) and the composition of solution, which is called Henry's law.

According to this law, "The mass of a gas dissolved in a given volume of the liquid at constant temperature is directly proportional to the pressure of the gas present in equilibrium with the liquid."

If the mass of gas is m which is dissolved in unit volume of solvent at pressure (P) then according to Henry law:-

$$m \propto P$$

$$m = kP \quad \text{———— (1)}$$

where, k = Henry constant

→ on the basis of equation (1), a straight line is obtained on drawing a graph b/w solubility of gases and equilibrium pressure.

When more than one gases together are come in contact with solvent then the amount of solubility of each gas is directly proportional to its partial pressure.

The amount of gas from all gases is determined by its mole fraction.

$$x \propto P_1 \quad \text{where } x = \text{mole fraction of gas} \\ P_1 = \text{Partial pressure of gas}$$

$$\text{or} \quad x = k' P_1 \\ P_1 = x/k'$$

$$\text{or} \quad P = kx \quad \left[k = \frac{1}{k'} \right]$$

→ At constant pressure, the volume of gas in solution is not affected by pressure.

Let m g of gas of molecular weight (M) at temperature (T) has volume (V). Gas is soluble in certain volume of solvent at equilibrium pressure (P). Then, ideal gas equation is.

$$\text{or} \quad PV = nRT \\ PV = \frac{m}{M} RT$$

$$\text{or} \quad V = \frac{m}{P} \cdot \frac{RT}{M}$$

→ According to Henry law, at constant temperature, $\frac{m}{P} = \text{constant}$, $V = \text{constant}$

Limitations of Henry law:

Henry law is applicable when:

- Pressure is not very high.
- Temperature is not very low.
- Gas does not react chemically with solvent and also does not associate and ionise.
- The solubility of gas is lesser in solvent means the solution

is dilute.

Vapour Pressure of Liquid Solutions:-

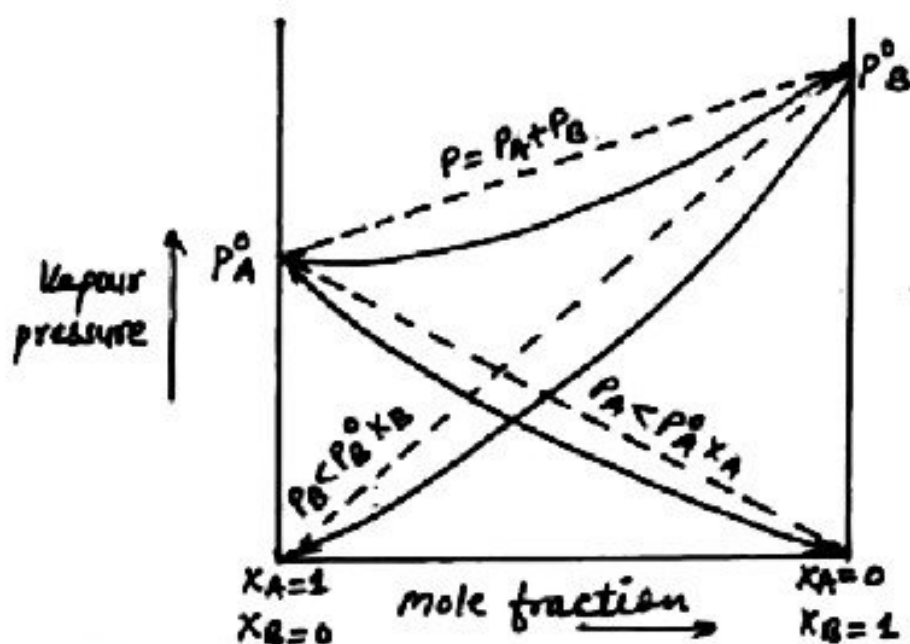
If a liquid is taken in an open container then the process of evaporation takes place to unlimited time because those particles of higher energy which go outside from the surface, most of them, escape in atmosphere due to open container.

But if a liquid is taken in closed container then the evaporation does not take place to unlimited time because, there is certain place for the movement of molecules of vapours. So these molecules are again mixed at the surface of liquid.

The process by which the molecules of vapours are mixed at the surface of liquid i.e. change of vapour to liquid state is known as condensation.

→ Evaporation and condensation are opposite reactions of each other. When the rate of evaporation is equal to the rate of condensation, then the equilibrium is established.

Evaporation $\rightleftharpoons$ Condensation



Vapour pressure curve showing negative deviation from Raoult's law

Raoult's Law as a special case of Henry's Law:-

According to Raoult's Law.

the Vapour pressure of a Volatile Component in a given solution is given by

$$P_i = x_i P_i^0$$

In the solution of a gas in a liquid, one of the components is so volatile that it exists as a gas and we have already seen that its solubility is given by Henry's law which states that -

$$[P = k_H x]$$

→ If we compare the equations for Raoult's law and Henry's law, it can be seen that the partial pressure of the Volatile component or gas is directly proportional to its mole fraction in solution. Only the proportionality constant k_H differs from P_i^0 .

Thus Raoult's law becomes a special case of Henry's law in which k_H becomes equal to P_i^0 .

Vapour pressure of solutions of solids in liquids:-

In this type of

solution, solution consists of solid dissolved in liquid.

e.g. Sodium chloride (NaCl), glucose, urea and cane sugar in water and Iodine and Sulphur dissolved in Carbon disulphide.

→ Some physical properties of these solutions are quite different from those of pure solvents. e.g. Vapour pressure.

Liquids at a given temperature vapourise and under equilibrium conditions, the pressure exerted by the vapour of the liquid over the liquid phase is called vapour pressure.

→ In a pure liquid the entire surface is occupied by the molecules of the liquid.

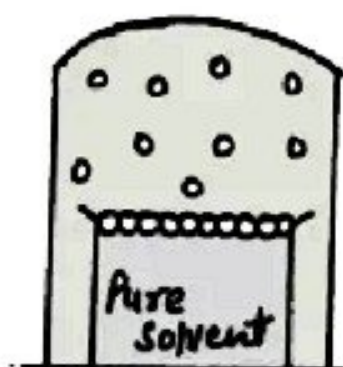
If a non-volatile solute is added to a solvent to give a solution, the vapour pressure of the solution is solely from the solvent alone.

This vapour pressure of the solution at a given temperature is found to be lower than the vapour pressure of the pure solvent.

at a same temp.

In the solution, the surface has both solute and solvent molecules, thereby the fraction of the surface covered by the solvent molecules gets reduced.

Consequently, the number of solvent molecules escaping from the surface is correspondingly reduced, thus the vapour pressure is also reduced.



(a)



(b) { $\circ$ -solvent $\bullet$ -solute }

Fig- Decrease in the vapour pressure of the solvent on account of the presence of solute in solvent.

(a) evaporation of the molecules of the solvent from its surface is denoted by $\circ$

(b) In a solution, solute particles have been denoted by $\bullet$ and they also occupy part of the surface area.

→ The decrease in the vapour pressure of solvent depends on the quantity of non-volatile solute present in the solution, irrespective of its nature.

Eg- decrease in the vapour pressure of water by adding 1.0 mol of sucrose to 1 kg. of water is nearly similar to that produced by adding 1.0 mol of urea to the same quantity of water at the same temperature.

→ Raoult's law in its general form can be stated as -

"For any solution, the partial vapour pressure of each volatile component in the solution is directly proportional to its mole fraction."

→ In a binary solution, let us denote the solvent by 1 and solute by 2. When the solute is non-volatile, only the solvent molecules are present in vapour phase and contribute to vapour pressure.

→ The vapour pressure of the solvent, x_1 be its mole fraction, P_1^0 be its vapour pressure in the pure state.

Then according to Raoult's law -

$$P_1 \propto x_1$$

$$\& P_1 = x_1 P_1^0$$

→ The proportionality constant is equal to the vapour pressure of pure solvent, P_1^0 .

A plot b/w the vapour pressure & the mole fraction of the solvent is linear

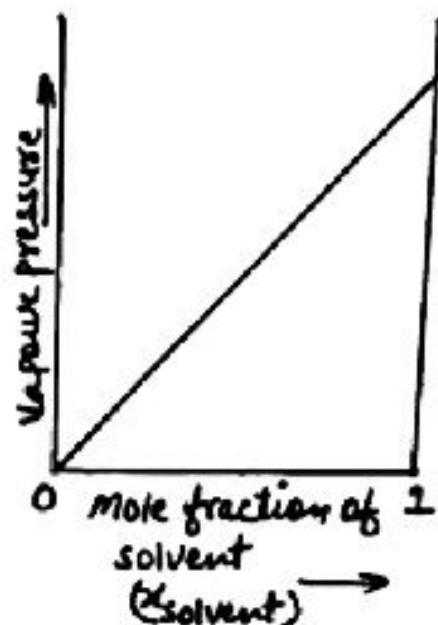


Fig → If a solution obeys Raoult's law for all conc., its vapour pressure would vary linearly from zero to the vapour pressure of the pure solvent.

Ideal Solution: → The solution which obey Raoult's law at the entire range of concentration and temperature is called ideal solution.

(i) It obeys Raoult's law.

$$\text{So, } P_A = P_A^0 x_A$$

$$P_B = P_B^0 x_B$$

$$P = P_A + P_B$$

$$P = P_A^0 x_A + P_B^0 x_B$$

(ii) The total volume of the solution is equal to the sum of volume of solute (B) and solvent (A).

$\Delta V_{\text{mixture}} = 0$ - i.e., change in volume is equal to zero.

(iii) Heat is neither absorbed nor emitted on mixing solute (B) and solvent (A).

$\Delta H(\text{mixing}) = 0$, i.e., the change in enthalpy is zero.

(iv) The magnitude of attractive force between A-B is equal to the sum of attractive forces present in A-A and B-B in solution.

(v) The solute and solvent having similar physical and chemical properties form ideal solution.

Example →

- (a) n-Hexane + n-Heptane (alkane)
- (b) Methanol + Ethanol (alcohol)
- (c) 1-chloro butane + 1-bromo butane (alkyl halide)
- (d) Benzene + Toluene (aromatic hydrocarbon)
- (e) CCl_4 + SiCl_4 (chloride of group 14)
- (f) Ethyl chloride + Ethyl bromide (alkyl halide)
- (g) chlorobenzene + Bromobenzene (aryl halide)

≠ Non-Ideal Solutions → Those solutions which do not obey Raoult's law and are accompanied by change in volume and change in enthalpy during their formation are called non-ideal solutions.

For non-ideal binary solution -

(i) $P_A \neq P_A^\circ x_A$ and $P_B \neq P_B^\circ x_B$

(ii) $\Delta V_{\text{mixing}} \neq 0$

(iii) $\Delta H_{\text{mixing}} \neq 0$

Non-ideal solutions are of two types:

(i) Non-ideal solutions with positive deviation:- In such type of solution, solvent-solvent (A-A) and solute-solute (B-B) inter molecular forces of attraction are stronger than solvent-solute (A-B) interaction.

For positive deviation →

(i) $P_A > P_A^\circ x_A$ and $P_B > P_B^\circ x_B$

(ii) $\Delta H_{\text{mixing}} = \text{Positive (endothermic)}$

(iii) ΔV_{mixing} = positive (high volume of solution)

→ There are following examples of some non-ideal solutions showing positive deviation:

- (i) Methanol + water
- (ii) Methanol + CCl_4
- (iii) Ethanol + water
- (iv) Acetone + Ethanol
- (v) Acetone + CS_2

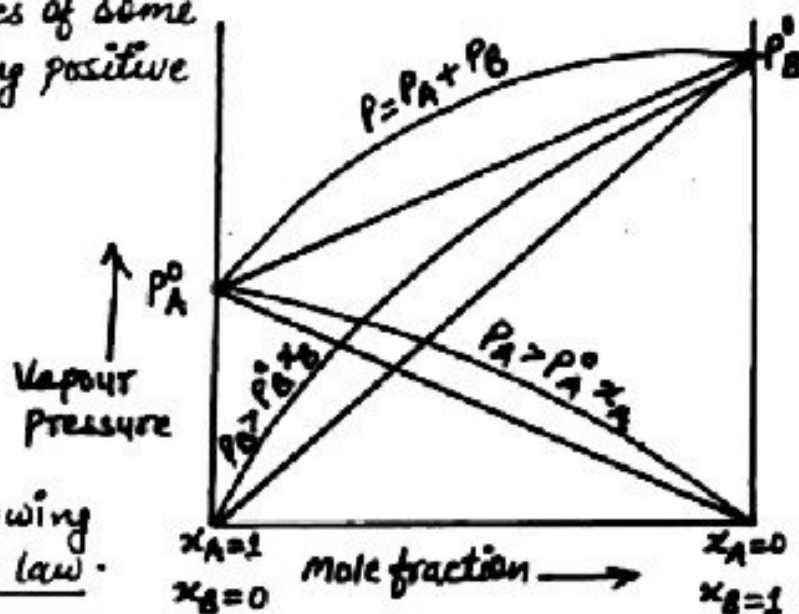


Fig - Vapour pressure Curve showing (+)ve deviation from Raoult's law.

(ii) Non-Ideal Solutions with negative deviation:-

In such type of solution, the partial vapour pressure of each component of solution is less than the vapour pressure as expected according to Raoult's law.

$$P_A < P_A^0 x_A$$

$$\& \quad P_B < P_B^0 x_B$$

For negative deviation,

ΔH_{mixing} = negative (exothermic)

ΔV_{mixing} = negative (low volume of solution)

→ There are following examples of non-ideal solution showing negative deviation:

- (i) CHCl_3 + Acetone
- (ii) water + HCl
- (iii) Acetone + Aniline
- (iv) Methanol + acetic acid
- (v) CHCl_3 + Benzene

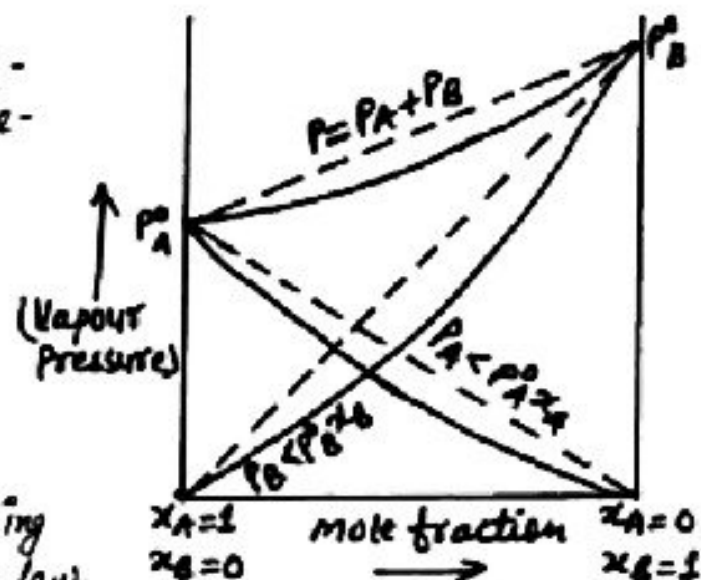


Fig: - Vapour pressure curve showing negative deviation from Raoult's law.

** Azeotropic Mixtures or constant Boiling Mixtures -

A mixture of two or more liquid components, having a definite composition in liquid and vapour phase and boil like a pure liquid at a particular temperature is called constant boiling mixture or azeotropic mixture. It is of two types.

⇒ Maximum Vaporisable and minimum boiling azeotropes →

In this type of mixture, the boiling point of azeotropic solution is lower than the boiling point of either of the pure components.

This is a non-ideal solution with positive deviation.

Example:- Water + Ethanal and Ethanol + Chloroform etc.

→ Minimum Vaporisable and maximum boiling azeotropes:-

→ In this type of mixture, the boiling point of azeotropic solution is higher than the boiling point of either of the pure component.

This is non-ideal solution with negative deviation.

Example - Acetone + Chloroform; Methanol + Acetic acid etc.

** Colligative properties and determination of Molar Mass →

Some properties of solution depend on the amount of solute dissolved in solution. These properties are not related to nature of solute and its composition. These properties of solution are called colligative properties. Some colligative properties of given below -

- (i) Relative lowering of vapour pressure.
- (ii) Elevation in boiling point.
- (iii) Depression in freezing point.
- (iv) Osmotic pressure.

(i) Relative Lowering in Vapour Pressure → The relative lowering in Vapour pressure of solution of non-volatile solid solute is equal to mole fraction of the solute.

$$\frac{P_A^\circ - P_A}{P_A^\circ} = x_B$$

$$x_B \frac{n_B}{n_A + n_B} = \frac{W_B/M_B}{W_A/M_A + W_B/M_B}$$

Here -

n_B = No. of moles of solute

n_A = No. of moles of solvent

W_B = Mass of solute

W_A = Mass of solvent

M_B = Molecular mass of solute

M_A = Molecular mass of solvent

According to Raoult's law,

$$x_B = \frac{P_A^\circ - P_A}{P_A^\circ}$$

$$\frac{W_B/M_B}{W_A/M_A + W_B/M_B} = \frac{P_A^\circ - P_A}{P_A^\circ}$$

For dilute solutions, $\frac{W_B}{M_B} \ll \frac{W_A}{M_A}$

So, $\frac{W_B}{M_B}$ in above equation can be assume negligible.

$$\therefore \frac{W_B/M_B}{W_A/M_A} = \frac{P_A^\circ - P_A}{P_A^\circ}$$

$$\frac{W_B \times M_A}{M_B \times W_A} = \frac{P_A^\circ - P_A}{P_A^\circ}$$

$$M_B = \frac{W_B \times M_A}{W_A} \left[\frac{P_A^\circ}{P_A^\circ - P_A} \right]$$

→ Molecular mass of non-volatile solute can be determined from above equation, if the values of all the others properties are known.

Example → Vapour pressure of pure benzene at a given temperature is 0.850 barr. 0.5 g non-volatile solute is dissolved in 39 g benzene. The Vapour pressure of solution obtained is 0.845 barr. Determine the molecular mass of solute (mol. mass of benzene = 78 g mol⁻¹)

Ans - $P_{C_6H_6}^0 = 0.850 \text{ barr}$

Vapour pressure of solution $P_A = 0.845 \text{ barr}$.

$W_B = 0.5 \text{ g}$; $M_{C_6H_6} = 78 \text{ g}$; $W_{C_6H_6} = 39 \text{ g}$; $M_B = ?$

$$M_B = \frac{W_B \times M_{C_6H_6}}{W_{C_6H_6}} \times \frac{P_{C_6H_6}^0}{P_{C_6H_6}^0 - P_A}$$

$$M_B = \frac{0.5 \times 78}{39} \times \frac{0.850}{0.850 - 0.845}$$

$$M_B = 170 \text{ gm.}$$

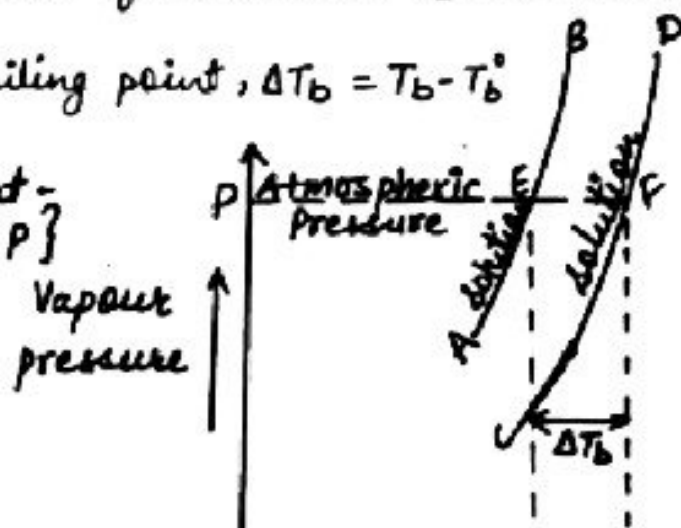
(ii) Elevation in boiling point :- The boiling point of a liquid is defined as the temperature at which its vapour pressure becomes equal to atmospheric pressure. We know that, the vapour pressure of solution having non-volatile solute is low.

Hence, more temperature will be required for the vapour pressure of solution to be equal to the atmospheric pressure, i.e., the boiling point of solution is more than boiling point of pure solvent.

→ If the boiling point of solution is T_b and boiling point of solvent is T_b^0 , then -

Elevation in boiling point, $\Delta T_b = T_b - T_b^0$

Figure shows that -
 $\{ T_b \propto \Delta P \}$



According to Raoult's Law, the lowering in vapour pressure of solute is proportional to the mole fraction.

$$\Delta P \propto x_B \quad (x_B = \text{mole fraction of solute})$$

$$\Delta T_b \propto x_B$$

$$\Delta T_b = K x_B$$

$$x_B = \frac{W_B/M_B}{W_A/M_A + W_B/M_B}$$

$$\Delta T_b = K \frac{W_B/M_B}{W_A/M_A + W_B/M_B}$$

For dilute solutions, $\frac{W_B}{M_B} \ll \frac{W_A}{M_A}$

$$\therefore \Delta T_b = K \frac{W_B/M_B}{W_A/M_A}$$

$$= K \cdot M_A \cdot \frac{W_B}{M_B \times W_A}$$

$$\boxed{\Delta T_b = K_b \cdot m}$$

Here, m = Molality of solution

K_b = Molal elevation constant

→ Relationship b/w elevation in boiling point and molecular mass of non-volatile solute —

We know that, $\Delta T_b = K_b \cdot m$

and molality of solution $m = \frac{W_B \times 1000}{M_B \times W_A}$

$$\therefore \Delta T_b = K_b \cdot \frac{W_B \times 1000}{M_B \times W_A}$$

$$\underline{\Delta T_b} \quad M_B = \frac{K_b}{\Delta T_b} \times \frac{W_B}{M_B} \times 1000 \quad \left\{ \text{Here, } W_B = \text{mass of solute} \right.$$

M_B = Molecular mass of solute

W_A = Mass of solvent

K_b = molal elevation constant }

→ According to Van't Hoff, K_b is related to latent heat of vaporisation of solvent as follows:

$$\left\{ K_b = \frac{M_A R T_b^2}{\Delta H_{\text{vaporisation}} \times 1000} \right\}$$

Here -

M_A = Molecular mass of solvent

R = Gas constant

T_b = Boiling point of solvent

($\Delta H_{\text{vaporisation}}$ = Latent heat of Vaporisation)

Example :-> The boiling point of benzene is 353.23 K. The boiling point of solution becomes 354.11 K on dissolving 1.80 g of volatile solute in 90 g benzene.

Calculate the molecular mass of solute. (K_b for benzene = $2.53 \text{ K kg mol}^{-1}$)

Ans -

$$M_B = \frac{K_b \times \frac{W_B}{W_A} \times 1000}{\Delta T_b}$$

Given, $K_b = 2.53 \text{ K kg mol}^{-1}$

$$\Delta T_b = 354.11 - 353.23$$

$$\Delta T_b = 0.88 \text{ K}$$

$$W_B = 1.80 \text{ g}$$

$$W_A = 90 \text{ g}$$

$$\therefore M_B = \frac{2.53}{0.88} \times \frac{1.80}{90} \times 1000 = 58 \text{ g mol}^{-1}$$

(iii) Depression in Freezing point :-> The temperature at which the vapour pressure of both solid and liquid state becomes equal, is called freezing point of liquid.

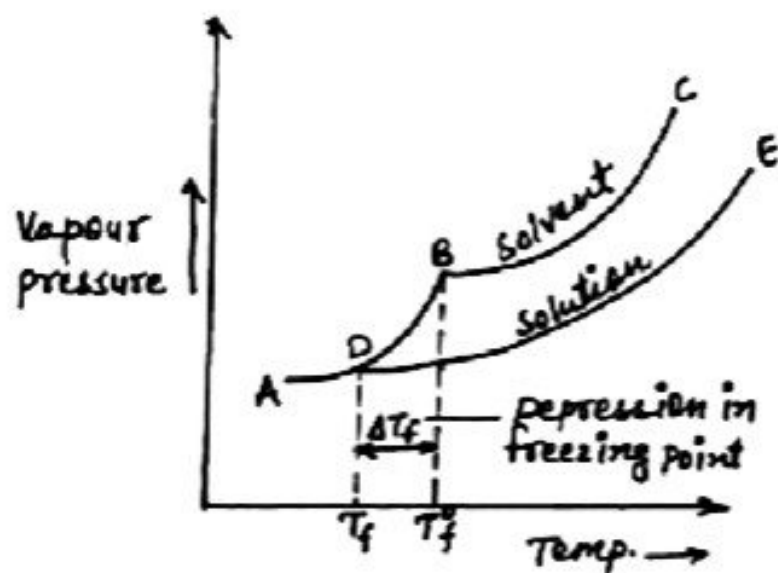
When non-volatile solute is dissolved in solvent, the vapour pressure of solution becomes lesser than solvent.

Hence, solution freezes at low temperature as compared to solvent, so, freezing point of solution becomes lesser than solvent.

→ If freezing point of pure solvent is T_f° and freezing point of solution is T_f , then -

Depression in freezing point -

$$\Delta T_f = T_f^\circ - T_f$$



→ In figure, Curve BC is vapour pressure of pure Solvent.

The solvent freezes in the form of solid at point B. The freezing point of solvent is represented as T_f° at point B. The vapour pressure of solution decreases on decreasing temperature which is shown by the curve DE.

The solution freezes at point D which represent the freezing point of solution (T_f).

So, according to figure, $\Delta T_f = T_f^\circ - T_f$ represents the depression in freezing point.

It is clear from the figure that, $\Delta T_f \propto \Delta P$

→ According to Raoult's law, the lowering in vapour pressure is proportional to mole fraction of solute -

So, $\Delta P \propto x_B$ (x_B - mole fraction of solute)

$$\Delta T_f = k x_B$$

$$x_B = \frac{W_B/M_B}{W_A/M_A + W_B/M_B}$$

$$\left\{ \Delta T_f = k \cdot \frac{W_B/M_B}{W_A/M_A + W_B/M_B} \right\}$$

for dilute solution - $\frac{W_B}{M_B} \ll \frac{W_A}{M_A}$

$$\therefore \Delta T_f = K \frac{W_B/M_B}{W_A/M_A}$$

$$\Delta T_f = K \cdot M_A \frac{W_B}{M_B \times W_A}$$

$$\boxed{\Delta T_f = K_f \cdot m}$$

Here - m = molality of solution

K_f = Molal depression constant or cryoscopic constant.

→ Relation b/w depression in freezing point and molecular mass of non-volatile solute →

We know that,

$$\Delta T_f = K_f \cdot m$$

and molality of solution -

$$m = \frac{W_B \times 1000}{M_B \times W_A}$$

$$\therefore \Delta T_f = K_f \frac{W_B \times 1000}{M_B \times W_A}$$

or

$$M_B = \frac{K_f}{\Delta T_f} \times \frac{W_B}{W_A} \times 1000$$

Here - W_B = mass of solute

M_B = Molecular mass of solute

W_A = Mass of solvent

K_f = molal depression constant. According to Van't Hoff, K_f is related to the latent heat of vaporisation of solvent as follows

$$\left\{ K_f = \frac{M_A R T_f^2}{\Delta H_{\text{vaporisation}} \times 1000} \right\}$$

Here - M_A = Molecular mass of solvent

T_f = Freezing point of solvent

R = Gas constant

$\Delta H_{\text{vaporisation}}$ = Latent heat of Vaporisation

Example - The depression in freezing point is found to be 0.68 K on dissolving 25.6 g sulphur in 1000 g naphthalene. Determine the molecular formula of sulphur.

(K_f for naphthalene = 6.8 K kg mol^{-1})

Ans.

$$M_B = \frac{K_f}{\Delta T_f} \times \frac{W_B}{W_A} \times 1000$$

Given $\rightarrow K_f = 6.8\text{ K kg mol}^{-1}$

$\Delta T_f = 0.68\text{ K}$

$W_B = 25.6\text{ g}$

$W_A = 1000\text{ g}$

$$M_B = \frac{6.8}{0.68} \times \frac{25.6}{1000} \times 1000$$

$$M_B = 256$$

$\rightarrow$ Let the molecular formula of sulphur is S_x -

$\therefore$ Molecular weight = $x \times 32$

$$x \times 32 = 256$$

$$x = \frac{256}{32} = 8$$

$\therefore$ The molecular formula of sulphur is S_8 .

Diffusion: $\rightarrow$ When potassium permanganate added in container of water then after some time violet colour spreads in container.

The process in which the movement of molecules takes place from high conc. area to low conc. area is known as diffusion.

Osmosis: $\rightarrow$ When thistle funnel having semi-permeable membrane filled with coloured solution of sugar is hanging in a container of water, then thistle funnel allows to pass molecules of solvent through it, but does not allow the molecules of solute.

The process of passing of solvent molecules from the solution of lower concentration to solution of higher concentration through a semi-permeable membrane is called osmosis.

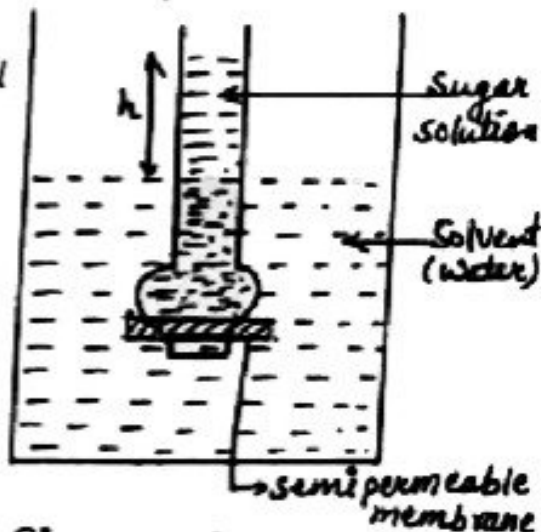


Fig - Osmis

→ Semi permeable membrane is made of chemical substance phenol or potassium ferrocyanide.

Semi permeable membrane is like a Continuous sheet which contains a network of very small pores which allow to pass molecules of solvent only.

(iv) Osmotic pressure → A rectangular apparatus is taken according to figure.

The apparatus consists of a chamber divided into 2 compartment by a semi-permeable membrane.

Both the compartments are fitted with pistons P_1 & P_2 . Two compartments A & B of the apparatus are filled with solvent & solution respectively.

The molecules of solvent move towards solution i.e., from A compartment to B compartment due to osmosis.

Due to this, the piston P_2 will be displaced upwards and P_1 will be displaced downwards.

To stop this movement of solvent, we have to apply mechanical pressure on solution side.

The pressure just sufficient to stop osmosis will be the osmotic pressure.

→ Hence the osmotic pressure may be defined as the excess pressure that must be applied to the solution side to prevent the passage of solvent into it through a semi-permeable membrane.

Osmotic pressure is represented by the symbol π .

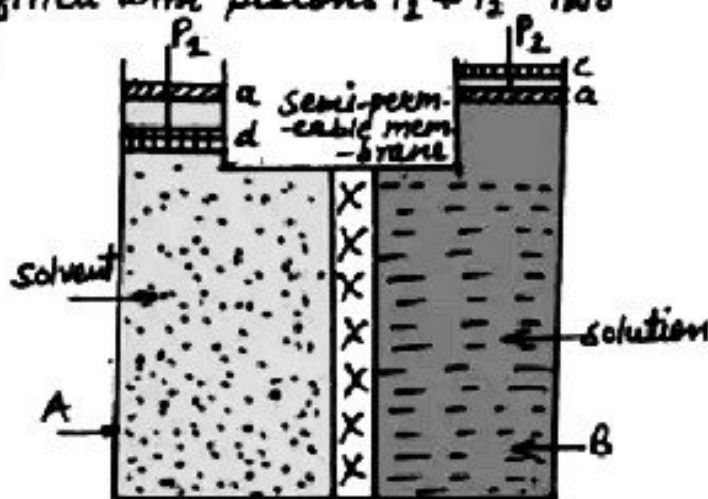


Fig-Osmotic pressure

* Rules for osmotic Pressure → Van't Hoff used gas laws on solution and derived a solution equation which is as follows:-

(i) Van't Hoff Boyle law: The osmotic pressure of a solution is directly proportional to its concentration (c) at a particular temperature.

$$\pi \propto c$$
$$\therefore \text{Concentration} = \frac{1}{\text{Dilution}(V)}$$

$$\therefore \pi \propto \frac{1}{V}$$

(ii) Van't Hoff pressure-Temperature law: The osmotic pressure of solution is directly proportional to its absolute temperature at a particular conc. and dilution.

$$\pi \propto T$$

On combination of above two laws-

$$\pi \propto \frac{T}{V}$$

$$\pi = \frac{ST}{V}$$

$$\pi V = ST \text{ (for 1 mole)}$$

$$\pi V = nST \text{ (for } n \text{ moles)}$$

Here S = solution constant whose value is equal to gas constant (R).

So, R is written in place of S
 $\{\pi V = nRT\}$

** Osmotic pressure and molecular mass of solute -

$$\pi V = nRT$$

$$\pi = \frac{n}{V} RT$$

$$\text{mole of solute } n = \frac{W_B}{M_B} \quad \left(\begin{array}{l} \text{Here - } W_B = \text{mass of solute} \\ M_B = \text{Molecular mass of solute} \end{array} \right)$$

$$\pi = \frac{W_B}{M_B V} RT$$

$$M_B = \frac{W_B RT}{\pi V}$$

Example:- Determine the osmotic pressure of $M/10$ solution of urea at 27°C temperature.

$$(R = 0.0821 \text{ atm L K}^{-1} \text{ mol}^{-1})$$

Ans - $\pi V = nRT$

$$\pi = \frac{n}{V} RT$$

$$\pi = CRT$$

Given - $C = \text{conc.} = \frac{1}{10} \text{ mole}$

$$R = 0.0821 \text{ atm L K}^{-1} \text{ mol}^{-1}$$

$$T = 27 + 273 = 300 \text{ K}$$

$$\pi = \frac{1}{10} \times 0.0821 \times 300 = 2.46 \text{ atm}$$

**** Isotonic Solutions**:- The two solution which can not allow the passage of solvent through semi-permeable membrane after separation, i.e., these have equal molar conc.

**** Hypertonic solutions**:- A solution having higher osmotic pressure than some other solution is said to be hypertonic with respect to the other solution.

**** Hypotonic solutions**:- A solution having lower osmotic pressure relative to some other solution is called hypotonic solution.

Example $\rightarrow$ 4% urea solution is isotonic to 12% solution of other organic compound A. Calculate the molecular mass of compound A.

Ans - $\text{osmotic pressure } (\pi) = \frac{WRT}{mV}$

$$\text{osmotic pressure of 4\% urea solution } (\pi_1) = \frac{4 \times R \times T}{60 \times 100} \times 1000$$

∴ The structural formula of urea is NH_2CONH_2 and molecular mass is 60 -

Osmotic pressure of 12% solution of compound A -

$$\pi_2 = \frac{12 \times R \times T}{M \times 100}$$

Isotonic solutions have equal osmotic pressure.

$$\pi_1 = \pi_2$$
$$\frac{4 \times R \times T}{6} = \frac{12 \times R \times T}{m}$$

$$m = \frac{72}{4}$$

$$m = 18 \text{ g}$$

(Molecular mass of compound A = 18 g)

Abnormal Molecular Mass :→ The molecular mass of solute is calculated from colligative properties accurately only when:

- (i) The solution is dilute, so that Raoult's law is obeyed.
- (ii) The solute neither undergoes dissociation nor association in solution.

If the dissociation or association of solute takes place on dissolving it in solvent, then the number of particles in solution increases or decreases and the molecular mass of solute is different from actual molecular mass due to change in colligative properties.

This molecular mass is called abnormal molecular mass.

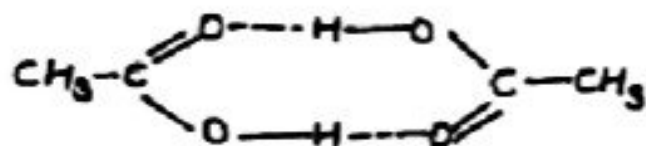
E.g:

2 mole of NaCl , CaCl_2 , $\text{K}_4[\text{Fe}(\text{CN})_6]$ dissolves in water and ionization is 100% then 1 mole of them form 2, 3, 5, mole ion respectively.

→ Since colligative property is inversely proportional to the molecular mass. Hence, the molecular masses of above compounds.

are obtained as $\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{5}$ part respectively from actual molecular mass.

So, formic acid, acetic acid, benzoic acid form dimers in benzene solution due to H-bonding.



Since, colligative property is inversely proportional to molecular mass. So, the molecular mass of carboxylic acids are obtained as twice the actual molecular mass.

→ Van't Hoff factor → Van't Hoff introduced a factor ' i ' to explain the abnormal molecular mass. This is called Van't Hoff's factor.

Van't Hoff's factor can be defined as follows-

$$i = \frac{\text{Observed value of colligative property}}{\text{Calculated value of colligative property}}$$

$$i = \frac{\text{No. of particles obtained after dissociation/association}}{\text{No. of particles taken before dissociation/association}}$$

$$i = \frac{\Delta P_{\text{observed}}}{\Delta P_{\text{calculated}}} = \frac{\Delta T_{\text{observed}}}{\Delta T_{\text{calculated}}} = \frac{\Delta T_f \text{ observed}}{\Delta T_f \text{ calculated}}$$

$$\left[i = \frac{\pi_{\text{observed}}}{\pi_{\text{calculated}}} \right]$$

→ If $i=1$, the solute does not dissociate or associate in solution.

$i > 1$, the solute is dissociated.

$i < 1$, the solute is associated.

→ The general equation of colligative properties show following changes due to dissociation or association of solute in solution:

(i) Relative lowering of vapour pressure $= \frac{P_A^0 - P_A}{P_A^0} = i x_B$

(ii) Elevation in boiling point $\Rightarrow \Delta T_b = i \cdot K_b \cdot m$

$$\Delta T_b = i \times K_b \times \frac{W_B \times 1000}{M_B \times W_A}$$

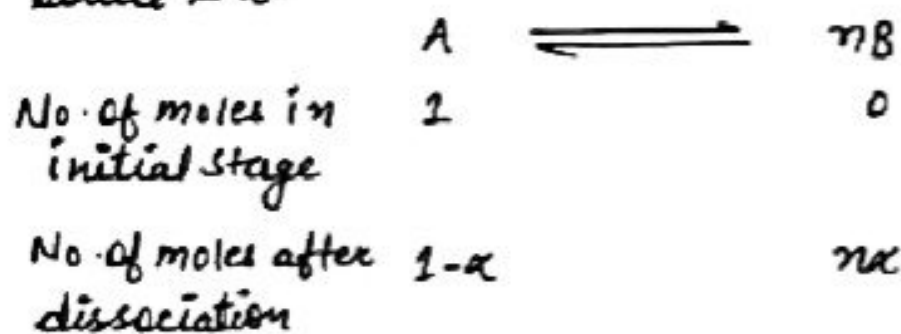
(iii) Depression in freezing point $\Rightarrow \Delta T_f = i \cdot K_f \cdot m$

$$\Delta T_f = i \cdot K_f \times \frac{W_B \times 1000}{M_B \times W_A}$$

(iv) osmotic pressure $\Rightarrow \pi = i C R T$
 $\pi = \frac{i \times W_B \times R \times T}{M_B \times V}$

* Relationship b/w Van't Hoff's factor (i) and degree of dissociation (α) $\Rightarrow$

Suppose A mole of solute is dissolved in solvent and dissociates to give n -particle. Then degree of dissociation of solute is α .



$$\begin{aligned} \text{No. of observe moles of solute} &= 1-\alpha + n\alpha \\ &= 1 + \alpha(n-1) \end{aligned}$$

$$\text{No. of theoretical moles of solute} = 1$$

$$i = \frac{\text{No. of moles after dissociation}}{\text{No. of moles before dissociation}}$$

$$i = \frac{1 + n(n-1)}{1}$$

$$\alpha = \frac{i-1}{n-1}$$

Example $\rightarrow$ 0.1 M solution of $K_4[Fe(CN)_6]$ was 46% ionised at $27^\circ C$ temperature. Calculate the osmotic pressure of solution.

$$(R = 0.082 \text{ atm L K}^{-1} \text{ mol}^{-1})$$

$$\pi = iCRT$$

Ans - Given van't Hoff's factor -

$$i = \frac{1 + \alpha(n-1)}{1}$$

$$\text{and } \alpha = \frac{46}{100} \quad (\text{for ionic compound } K_4[Fe(CN)_6])$$

$$n = 5$$

$$\Rightarrow i = 1 + \frac{46}{100}(5-1) = 2.84$$

$$C = 0.1, R = 0.082, T = 27 + 273 = 300 K$$

$$\pi = iCRT = 2.84 \times 0.1 \times 0.082 \times 300 = 6.98 \text{ atm.}$$

$\rightarrow$ Relationship b/w van't Hoff's factor and degree of association (α) $\rightarrow$

Let n molecules of solute associate to form a molecule and degree of association is α .



Initial No. of mole 1 0
No. of moles after $(1-\alpha)$ α/n
association

$$\text{No. of observed moles of solute} = 1 - \alpha + \frac{\alpha}{n}$$

$$\text{No. of theoretical moles of solute} = 1$$

$$i = \frac{\text{No. of moles after association}}{\text{No. of moles before association}}$$

$$\Rightarrow i = \frac{1 - \alpha + \frac{\alpha}{n}}{1}$$

$$\Rightarrow i = 1 - \alpha + \frac{\alpha}{n}$$

$$\Rightarrow i n = n - n \alpha + \alpha$$

$$\Rightarrow i n - n = \alpha - n \alpha$$

$$\Rightarrow n(i - 1) = \alpha(1 - n)$$

$$\Rightarrow \boxed{\alpha = \frac{n(i - 1)}{1 - n}}$$

Example $\rightarrow$ 0.3 g benzoic acid is dissolved in 20 g benzene. The depression in freezing point of this solution is 0.317 K. Calculate the degree of association of benzoic acid.
(K_f for benzene = $5.1 \text{ K kg mol}^{-1}$)

Ans - $M_B \text{ observed} = \frac{K_f}{\Delta T_f} \times \frac{W_B}{W_A} \times 1000$

Given - $K_f = 5.1 \text{ kg mol}^{-1}$, $\Delta T_f = 0.317 \text{ K}$, $W_B = 0.3 \text{ g}$, $W_A = 20 \text{ g}$

$$M_B \text{ observed} = \frac{5.1 \times 0.3 \times 1000}{0.317 \times 20}$$

$$M_B \text{ observed} = 241$$

$$i = \frac{\text{theoretical molecular mass (of benzoic acid)}}{\text{observed molecular mass}}$$

$$\Rightarrow i = \frac{122}{241} = 0.506$$

Benzoic acid forms dimer in benzene, so $n = 2$ -

$$\alpha = \frac{n(i - 1)}{1 - n}$$

$$\alpha = \frac{2(0.506-1)}{1-2}$$

$$\alpha = \frac{2 \times (-0.494)}{-1}$$

$$\alpha = 0.988$$

$$\alpha = 98.8\%$$
